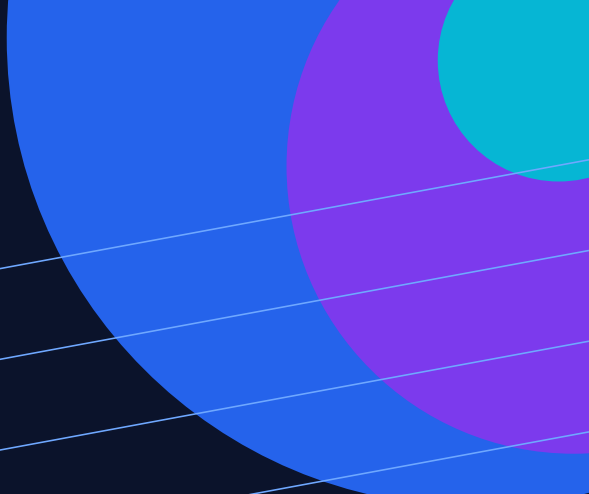




THREAT RESEARCH / DETECTION ENGINEERING

HOLLOWGRAPH Calendar C2

Exact-date Microsoft Graph detection candidate

REPORT	STATUS	CONFIDENCE
TECHNICAL THREAT STUDY	PRODUCTION CANDIDATE	HIGH - SOURCE & TELEMETRY

Prepared for defensive research and tenant-specific validation. This report separates source-observed facts from analytical interpretation and proposed detection logic.

VERSION 1.0 / 22 JULY 2026

Threat Overview

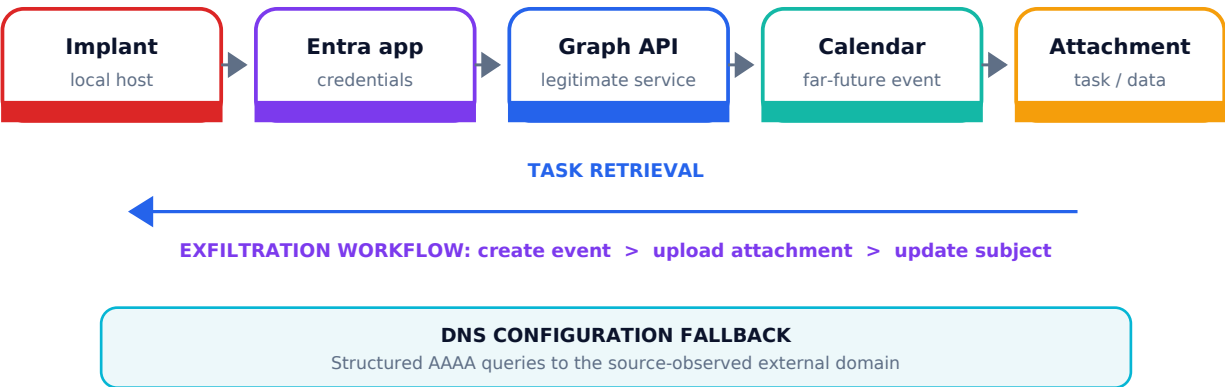
**WHY THIS
MATTERS**

HOLLOWGRAPH demonstrates how a compromised Microsoft 365 identity and a legitimate SaaS API may be repurposed as a covert two-way channel. Destination-based controls may see ordinary Microsoft Graph traffic while the operational meaning is hidden in calendar objects.

Source-observed behavior

Stage	Observed fact	Defensive relevance
Configuration	The analyzed .NET implant used Entra ID application credentials stored locally in logAzure.txt.	A local credential artifact and application identity provide endpoint and identity investigation pivots.
Task retrieval	The implant issued GET requests to /users/{mailbox}/calendarView for 13 May 2050 and read tasking from an attached file.	The endpoint and far-future date are present in the request URI and may be observable in Graph Activity Logs.
Exfiltration	The implant created an event, uploaded encrypted data as attachments, and changed the event subject.	The sequence is source-supported, but body content needed for specificity is not documented in Graph Activity Logs.
Credential refresh	Structured AAAA queries used a source-observed external domain to update configuration values.	Useful for IOC-led hunting and DNS investigation; insufficient alone for a durable generic rule.

Attack-channel model



Evidence Boundaries

Evidence class	Statement	Use in this package
Source fact	The analyzed sample queried calendarView with a date range on 2050-05-13.	Directly supports the exact-date production candidate.
Documented telemetry	MicrosoftGraphActivityLogs documents request method, URI, response status and identity/application context.	Supports filters and investigation fields without assuming body visibility.
Analyst interpretation	Far-future calendar access may be anomalous and may generalize beyond the exact sample.	Supports hunting only; maliciousness is not established.
Unvalidated assumption	The tenant preserves startDateTime in RequestUri without truncation or transformation.	Must be tested before deployment or promotion.

REQUEST METADATA

method, URI, status

VISIBLE

IDENTITY CONTEXT

app, SP, user, IP

VISIBLE

REQUEST BODY

subject, event date

NOT DOCUMENTED

ATTACHMENT BODY

name and content

NOT DOCUMENTED

SAFE DETECTION BOUNDARY

Use RequestMethod + RequestUri + ResponseStatusCode and documented identity fields.

EXCLUDED FROM THE RULE

Event subject, attachment filename, attachment type and payload content.

CORE CONTROL

No query in this package infers calendar subject, attachment filename, content type, event body or payload content. Those values are excluded unless a separate, validated telemetry source exposes them.

Detection Design

HIGH CONFIDENCE

TENANT TEST REQUIRED

EXACT BEHAVIOR

Detection hypothesis

IF

Microsoft Graph accepts a GET request to the calendarView endpoint and the request URI contains the exact 2050-05-13 date observed in the analyzed HOLLOWGRAPH sample, then the activity warrants investigation as a potential calendar-based dead-drop access.

Condition-to-evidence traceability

Condition	Reason	Evidence strength
RequestMethod == GET	Matches source-observed task retrieval.	High
RequestUri contains /calendarView	Limits activity to the source-observed Graph endpoint.	High
RequestUri contains 2050-05-13	Matches the exact far-future date used by the analyzed sample.	High but brittle
ResponseStatusCode 200-299	Retains requests accepted by Microsoft Graph.	High
No threshold	The source provides no basis for an arbitrary request count.	Conservative design

KQL implementation

```
MicrosoftGraphActivityLogs
| where TimeGenerated >= ago(1h)
| where RequestMethod == "GET"
| where RequestUri contains "/calendarView"
| where RequestUri contains "2050-05-13"
| where ResponseStatusCode between (200 .. 299)
| project TimeGenerated, AppId, ServicePrincipalId, UserId,
  ClientAuthMethod, IPAddress, UserAgent,
  RequestMethod, RequestUri, ResponseStatusCode
```

Telemetry & Investigation

REQUIRED SOURCE

Native Log Analytics table: MicrosoftGraphActivityLogs. Deployment must stop if the table is unavailable or RequestUri does not retain the required date parameter.

Field	Role in rule or triage	Required quality check
TimeGenerated	Scheduling and incident timeline.	Current ingestion and expected latency.
RequestMethod	Direct GET filter.	Consistent casing and non-null values.
RequestUri	Endpoint and exact-date detection.	Full query parameter retention; no truncation.
ResponseStatusCode	Accepted-request filter.	Numeric values populated consistently.
AppId / ServicePrincipalId	Application attribution and owner validation.	Resolvable to enterprise application inventory.
UserId / ClientAuthMethod	Delegated vs application-context analysis where supported.	Interpret only after tenant validation.
IPAddress / UserAgent	Enrichment and historical comparison.	Do not use as broad exclusions.

Recommended investigation pivots

Priority	Analyst question	Evidence to collect
1	Is the application approved and owned?	Enterprise app owner, publisher, permissions, creation date and change history.
2	Was the request expected?	Business purpose, application schedule, mailbox target and historical activity.
3	Does endpoint evidence support compromise?	Process lineage, local credential artifacts, Graph-bound connections and file activity.
4	Is there related identity abuse?	Service principal sign-ins, consent changes, credential additions and anomalous source context.

False Positives & Limitations

Legitimate scenario	Likelihood	How to validate	Tuning posture
Application test using extreme dates	Medium	Confirm owner, test window and non-production scope.	Time-bound exclusion only if documented.
Calendar migration or archival tooling	Low-Medium	Review product function and recurring query pattern.	Exclude stable approved Appld, not IP alone.
Long-range scheduling service	Low	Confirm business workflow and mailbox scope.	Narrow identity-based exclusion.
Security or QA testing	Medium	Verify change record and authorized operator.	Prefer maintenance-window suppression.

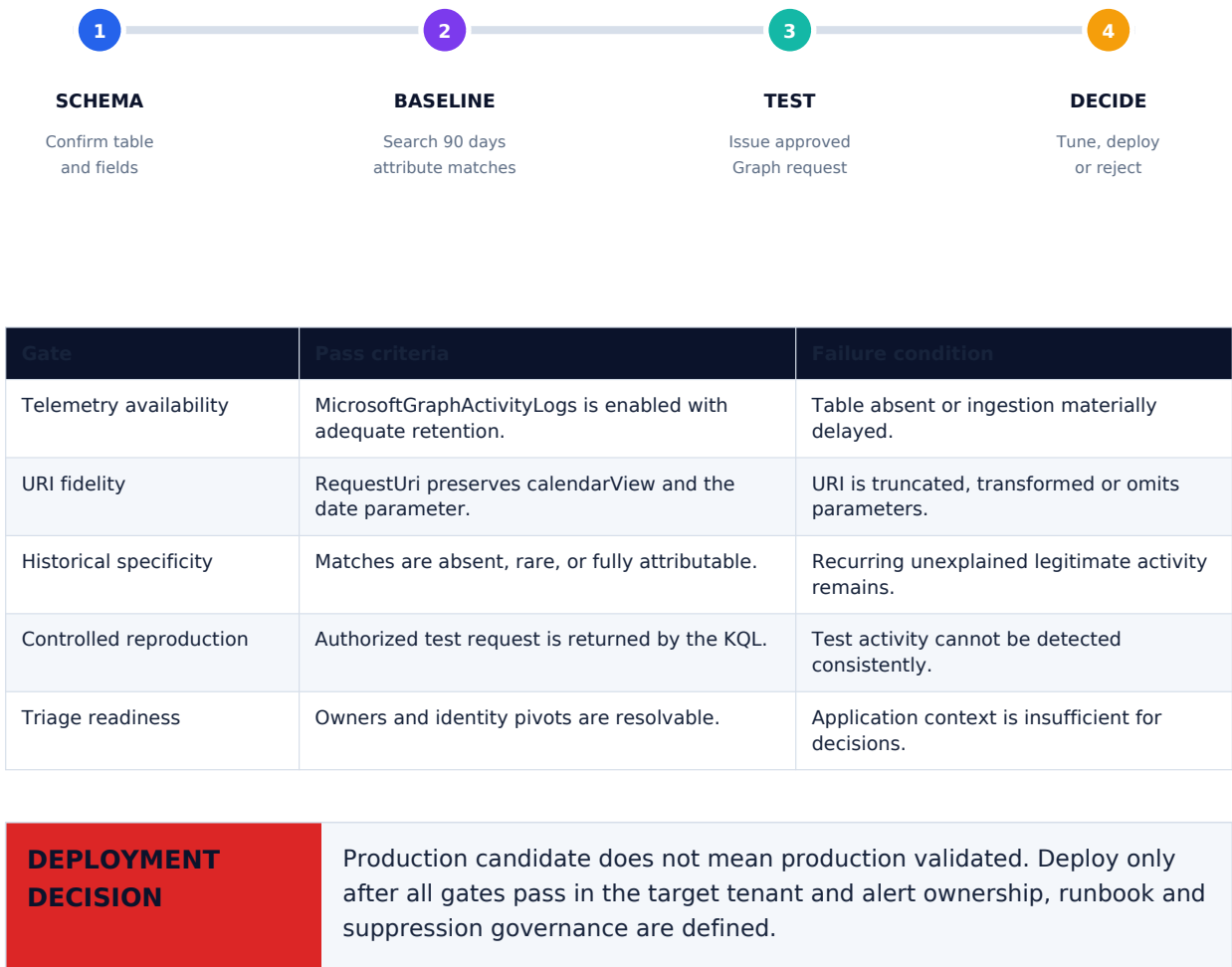
DURABILITY LIMIT

The candidate is precise because it matches the exact sample date. It is also brittle: changing the date, endpoint workflow or channel removes the match. The companion far-future hunt provides broader coverage but is not yet a production rule.

Explicit non-goals

Not detected	Reason
Event subject patterns	Subject content is expected in the request body and is not a documented Graph Activity Log field.
Attachment filename or content type	Attachment metadata/body visibility is not established in the required telemetry.
Encrypted exfiltrated content	Payload content is not available to this rule.
Date-shifted variants	The exact-date condition intentionally prioritizes confidence over breadth.

Validation & Deployment Gate



ATT&CK; Mapping & References

Technique	Relationship to observed behavior	Mapping posture
T1102.002 - Web Service: Bidirectional Communication	Microsoft Graph and calendar objects are used as a two-way dead-drop channel.	Direct behavioral relationship.
T1071.001 - Application Layer Protocol: Web Protocols	The implant communicates with Microsoft Graph over web protocols.	Supporting protocol relationship.

Primary threat research

1. Group-IB. HOLLOWGRAPH: Turning Microsoft Graph into a Covert C2 Channel.
group-ib.com/blog/hollowgraph-microsoft-365/

Platform and schema documentation

2. Microsoft Learn. Microsoft Graph activity logs overview.
learn.microsoft.com/en-us/graph/microsoft-graph-activity-logs-overview
3. Microsoft Learn. MicrosoftGraphActivityLogs table reference.
learn.microsoft.com/en-us/azure/azure-monitor/reference/tables/microsoftgraphactivitylogs

MITRE ATT&CK;

4. MITRE ATT&CK; T1102.002 - Web Service: Bidirectional Communication.
attack.mitre.org/techniques/T1102/002/
5. MITRE ATT&CK; T1071.001 - Application Layer Protocol: Web Protocols.
attack.mitre.org/techniques/T1071/001/

ANALYTICAL POSTURE

High confidence that the exact source-observed request can be represented with documented Graph Activity Log fields. Confidence does not extend to tenant prevalence, false-positive rate or production readiness until validation is complete.

Document controls

Field	Value
Research date	22 July 2026
Version	1.0
Audience	SOC analysts, detection engineers, threat hunters
Data handling	No customer data or internal indicators included
Review trigger	New source evidence, schema change, tenant test results or material false-positive findings